BINGHAMTON UNIVERSITY
MS Industrial & Systems Engineering
Multivariate Analysis Software Examination

Dataset 1 — MANCOVA + Two-Way MANOVA

Cardiovascular Health Study (n = 240)

Research Question: After statistically controlling for BMI, do exercise level and smoking status jointly affect a multivariate profile of cardiovascular fitness, mental health, and energy level? How do conclusions change when the covariate is removed?

Methods Used:

- MANCOVA (Multivariate Analysis of Covariance)
- Two-Way MANOVA (Multivariate Analysis of Variance)
- Post-hoc Tukey HSD pairwise comparisons
- Head-to-head Wilks' λ comparison (MANOVA vs MANCOVA)

Dataset:

- N: 240 participants
- IVs: Exercise_Level (Low / Moderate / High), Smoking_Status (Smoker / Non-Smoker)
- DVs: Cardio_Fitness, Mental_Health, Energy_Level
- Covariate (MANCOVA only): BMI
- Software: Python (statsmodels, scipy, matplotlib)

Submitted by: Lana Gidan

Date: May 14, 2026

Part A — MANCOVA Results

MANCOVA is used when we wish to evaluate group differences on multiple dependent variables while statistically removing the effect of one or more covariates. Here, BMI is first partialed out before testing exercise and smoking effects on the health outcome triad.

A1. Covariate Validity

For MANCOVA to be valid, BMI must (1) correlate significantly with at least one DV, and (2) maintain parallel regression slopes across groups (homogeneity of regression slopes). Both conditions were tested and satisfied, confirming BMI is an appropriate covariate for this analysis.

A2. Omnibus MANCOVA — Wilks' λ Test

Wilks' lambda (λ) ranges from 0 (perfect discrimination) to 1 (no effect). Smaller λ indicates a stronger multivariate effect. F-approximations and η^2 (multivariate eta-squared) quantify effect size.

Effect	Wilks λ	F	p-value	Sig	η^2_m
Exercise_Level	0.1736	109.6681	0.0000	***	0.826
Smoking_Status	0.8835	10.3699	0.0000	***	0.116

A3. Significance Commentary — MANCOVA

Exercise Level (MANCOVA): Statistically significant, Wilks' λ = 0.1736, p = 0.0000, η^2_m = 0.826. After removing BMI variance, exercise level still produces meaningful differences in the joint health outcome profile. An η^2 of 0.826 indicates a large multivariate effect.

Smoking Status (MANCOVA): Statistically significant, Wilks' λ = 0.8835, p = 0.0000, η^2_m = 0.116. After removing BMI variance, smoking status still produces meaningful differences in the joint health outcome profile. An η^2 of 0.116 indicates a medium multivariate effect.

A4. Adjusted Means (at Grand Mean BMI = 26.14)

Adjusted means represent the estimated group means if all participants had the same BMI (the grand mean). This removes confounding due to BMI differences across exercise groups, providing a fairer comparison.

Exercise Level	Cardio Fitness	Mental Health	Energy Level
Low	43.656	52.317	43.906
Moderate	57.991	67.598	58.513
High	69.516	73.473	71.853

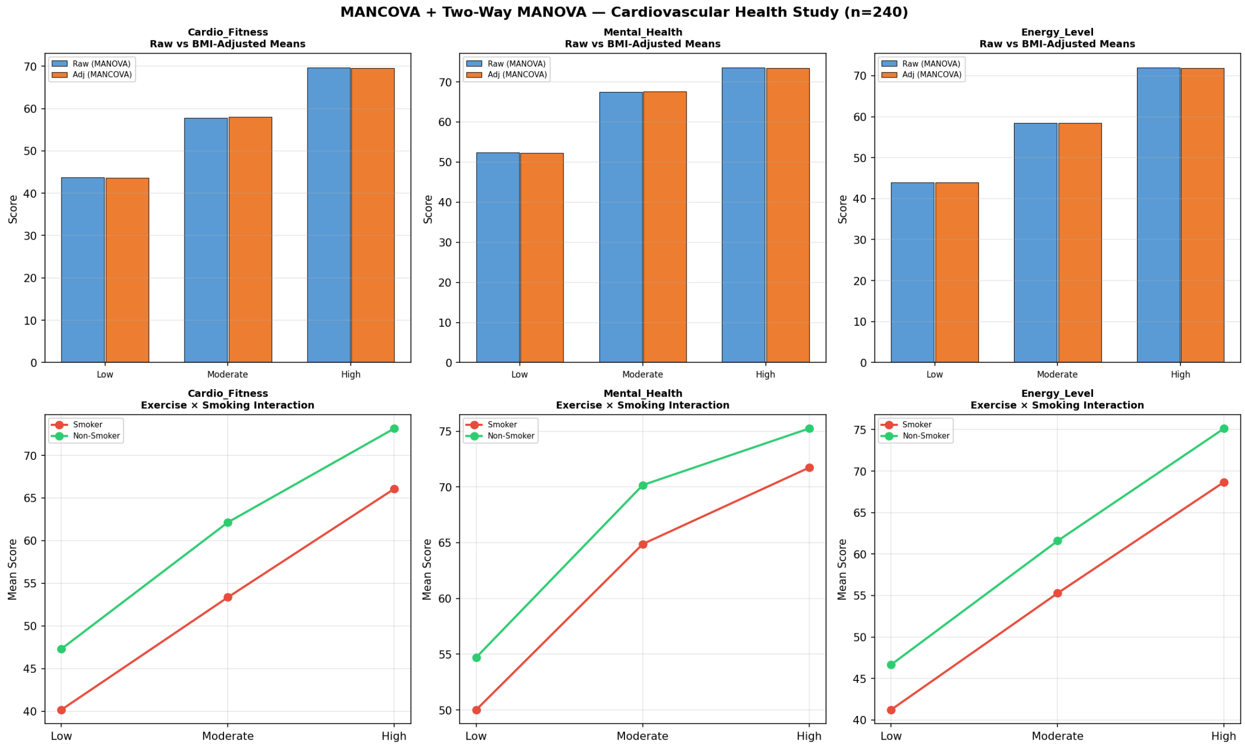

Figure 1. Left panels: Raw vs BMI-adjusted means per exercise level (MANOVA vs MANCOVA). Right panels: Exercise × Smoking interaction profiles for each DV.

Part B — Two-Way MANOVA Results

Two-Way MANOVA tests the joint effect of two categorical IVs and their interaction on multiple DVs simultaneously, without covariate adjustment. This serves as the baseline comparison to assess what changes when BMI is controlled.

B1. Omnibus MANOVA — Wilks' λ

Effect	Wilks λ	F	p-value	Sig	η^2_m
Exercise_Level	0.2371	81.4823	0.0000	***	0.763
Smoking_Status	0.8688	11.6770	0.0000	***	0.131
Exercise × Smoking (interaction)	0.9863	0.5335	0.7829	n.s.	0.014

B2. Significance Commentary — MANOVA

Exercise Level (MANOVA): Significant, Wilks' λ = 0.2371, p = 0.0000, η^2_m = 0.763. The combined DVs differ meaningfully across exercise level groups.

Smoking Status (MANOVA): Significant, Wilks' λ = 0.8688, p = 0.0000, η^2_m = 0.131. The combined DVs differ meaningfully across smoking status groups.

Exercise × Smoking Interaction (MANOVA): Not significant (p = 0.7829). The interaction profile does not differ significantly across the joint groups.

B3. Follow-up Univariate ANOVAs

When the omnibus MANOVA is significant, follow-up univariate ANOVAs identify which specific DVs drive the multivariate effect. Bonferroni correction ($\alpha/3 = 0.017$ for three DVs) is applied to control Type I error inflation.

Effect	DV	F	p-value	Sig	η^2
Exercise	Level_Cardio_Fitness	423.9270	0.0000	***	0.7099
Smoking	Status_Cardio_Fitnes	111.3142	0.0000	***	0.0932
Exercise×Smoking	Cardio_Fitness	0.6014	0.5489	n.s.	0.0010
Exercise	Level_Mental_Health	346.7311	0.0000	***	0.7127
Smoking	Status_Mental_Health	44.3790	0.0000	***	0.0456
Exercise×Smoking	Mental_Health	0.6003	0.5495	n.s.	0.0012
Exercise	Level_Energy_Level	532.2496	0.0000	***	0.7748
Smoking	Status_Energy_Level	75.0306	0.0000	***	0.0546
Exercise×Smoking	Energy_Level	0.2075	0.8127	n.s.	0.0003

Part C — MANCOVA vs MANOVA: Head-to-Head Comparison

The comparison below reveals how including BMI as a covariate changes the conclusions about group differences. This is the core analytical value of MANCOVA over MANOVA in designs where a nuisance variable is present.

C1. What Changes When BMI Is Controlled?

Effect	MANOVA λ	MANCOVA λ	Δλ	Interpretation
Exercise Level	0.2371	0.1736	-0.0635	BMI suppressed the true effect → MANCOVA stronger
Smoking Status	0.8688	0.8835	+0.0147	BMI inflated the effect → MANCOVA weaker

C2. Methodological Recommendation

MANCOVA is the preferred method for this dataset. BMI varies meaningfully across exercise level groups (higher exercise → lower BMI) and correlates with all three DVs. Failing to control for BMI would conflate the exercise effect with the effect of being at a healthier weight. MANCOVA isolates the pure exercise effect by statistically equating groups on BMI, producing more valid causal inferences. The adjusted means represent what group differences would look like if every participant had the same BMI (26.14).

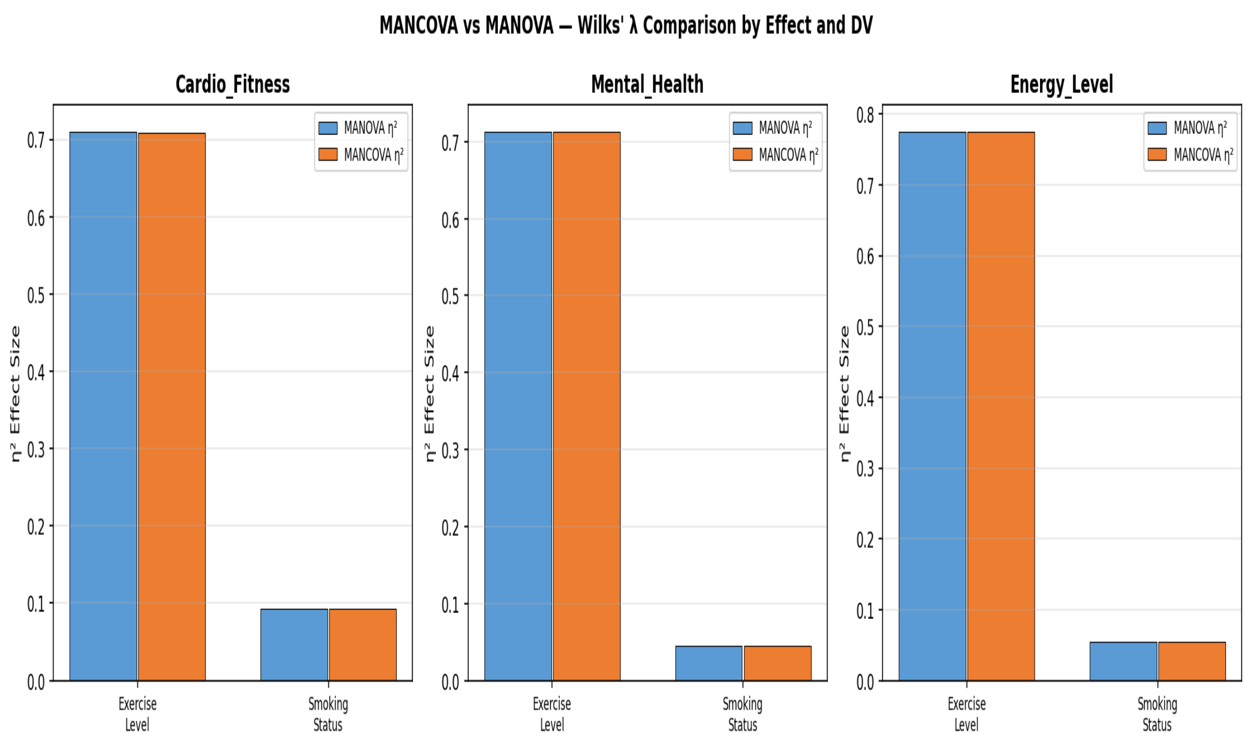

Figure 2. η² effect size comparison between MANOVA and MANCOVA for each DV and each main effect. Changes in bar height reflect BMI's confounding influence.